Question		Answer	Marks	AO	Guidance
5		$1 + 3 + 5 + \ldots + (2n \pm 1)$	M1	2.1	or identify AP with n terms and $a = 1$, $d = 2$; may be implied by award of first **A1**
		$\frac{n}{2}(2 \times 1 + (n-1) \times 2)$ **or** $\frac{n}{2}(1 + (2n-1))$	A1	1.1	use of formula for sum of arithmetic progression with n terms and $a = 1$ and $l = 2n - 1$; **or** $a = 1$ and $d = 2$; allow $\frac{n}{2}(2 + 2n - 2)$ but no further contraction
		$\frac{n}{2} \times 2n = n^2$ [which is a square number]	A1	2.2a	correct intermediate step needed for **A1**
		Alternative method (Further Maths 1)			
		$2\sum_{r=1}^{n} r - n$	M1		from $\sum_{r=1}^{n}(2r-1)$; allow for $\sum_{r=1}^{n}(2r+1)$; may be implied by award of first **A1**
		$=2 \times \frac{n(n+1)}{2} - n$	A1		all correct;
		$n^2 + n - n = n^2$ [which is a square number]	A1		correct intermediate step needed for **A1**
		Alternative method (Further Maths 2)			
		assume $\sum_{r=1}^{k}(2r-1) = k^2$ (i) working towards $\sum_{r=1}^{k+1}(2r-1) = (k+1)^2$ (ii)	M1		
		$\sum_{r=1}^{k}(2r-1) + (2(k+1)-1)$ $= k^2 + 2k + 1$ $= (k+1)^2$ so true for $n = k + 1$ if true for $n = k$	A1		correct intermediate step **and** statement needed
		$2 \times 1 - 1 = 1^2$ so true for $n = 1$ so by induction true for $n = 2, 3, 4, \ldots$	A1		
			[3]		